COMP3438 System Programming

Overview of Unix Demo

Tasks

Task 1: Accessing department Linux computer (e.g. apollo2) with your student account

Task 2: Basic Unix Skills

`ls`, `cd`, `cp`, `mv`, `rm`, `mkdir`, `who`, `whoami`, `gcc`, `cat`, `man`, `hexedit`, `wc`

Task 3: Text Editing

`vi`, `gedit`

Task 4: Shell output redirection e.g. `who > log.txt`, `who >> log.txt`

Task 5: Shell input redirection, e.g. `wc -l < log.txt`
Shell input and output redirection, e.g. `wc -l < log.txt > log2.txt`
Shell piping, e.g. `ls -l | wc -l > log3.txt`

Task 6: Write, compile, and execute a C program

Task 7: Install Linux (Ubuntu 32 bit) on VirtualBox using the prepared image

Task 8: Install Linux (e.g. kernel version 4.4.231, i.e. Ubuntu 16.04 LTS) on VirtualBox using image downloaded from Ubuntu website https://ubuntu.com

Task 9: Download Linux kernel source code (e.g. version 4.4.231) from http://kernel.org

Tasks

Task 10: Write a shell program that tests if a given file exists, and prints its numer of lines. Use the `$1` shell parameter, `test -f [file name]`, `wc -l`, and `echo` command.

Task 11: List all regular files in a directory, using `find` command.

Task 12: Print the number of lines of the files listed in Task 11, using ` ` embedding.

Task 13: List all .c files containing '`printf`' string using the `grep` command, and print the string's 2 lines contexts using `grep -C`.